X: RV

7/10/25 $E[X^n]$: n-th moment $n=1,2,\dots$

Notation: $\mu_X = E[X]$ mean, expected value

$Var(X) = E[(X-\mu_X)^2]$

$E[(X-\mu_X)^n]$: n-th centralized moment

$Y = \frac{X-\mu_X}{\sigma_X}$ $E[Y]=0$ "centralized"

$Var(Y)=1$ "normalized"

Moment generating function: (MGF)

X: RV

$t \in \mathbb{R}$

$M_X(t) \triangleq E[e^{tX}]$ (only for t for which expectation exists)

$M_X(0)=1$ Typical: t: interval around '0' $M_X(t)$ exists.

→ X: bounded $\Rightarrow M_X(t)$: exists for all t

Ex: ① $X \sim Bernoulli(p) \sim \{\underset{0}{1-p}, \underset{1}{p}\}$

$M_X(t) = E[e^{tX}] = (1-p) + (p)e^t = 1 + p(e^t-1)$

② $X \sim \{\underset{-2}{1/8}, \underset{-1}{1/8}, \underset{0}{1/2}, \underset{1}{1/8}, \underset{2}{1/8}\}$

$M_X(t) = \frac{1}{8}\cdot e^{-2t} + \frac{1}{8}e^{-t} + \frac{1}{2}\cdot 1 + \frac{1}{8}e^{t} + \frac{1}{8}e^{2t}$

③ $X \sim Geometric(p) \sim \{\underset{(1-p)^{k-1}\cdot p}{1}, 2, 3, \dots, k, \dots\}$

$M_X(t) = \sum_{k=1}^{\infty} (1-p)^{k-1} p \cdot e^{kt} = \frac{pe^t}{1-(1-p)e^t}$ if $(1-p)e^t < 1$

$t < \log \frac{1}{1-p}$

④ $X \sim Poisson(\lambda) \sim \{\underset{e^{-\lambda}\frac{\lambda^k}{k!}}{0}, 1, 2, \dots, k, \dots\}$

$M_X(t) = \sum_{k=0}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!}\cdot e^{kt} = e^{-\lambda}\cdot e^{\lambda e^t} = e^{\lambda(e^t-1)}$ $\forall t$.

⑤ $X \sim \text{Binomial}(n,p)$

$$M_X(t) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} \cdot e^{kt}$$

$$= \underbrace{\left(1-p+pe^t\right)^n}_{M_{\text{Ber}(p)}(t)}$$

Property: $X_1, X_2, \dots, X_n$: indep.

$$M_{X_1+\cdots+X_n}(t) = E\left[e^{t(X_1+\cdots+X_n)}\right] = M_{X_1}(t) M_{X_2}(t) \cdots M_{X_n}(t)$$

Ex: ① $X \sim \text{Uniform}[0,1]$

$$M_X(t) = E\left[e^{tX}\right] = \int_0^1 1 \cdot e^{tx}\, dx = \frac{e^t - 1}{t} \quad \text{if } t$$

② $X \sim$ pdf $\lambda e^{-\lambda x}$, $x > 0$

$$M_X(t) = \int_0^\infty \lambda e^{-\lambda x} \cdot e^{tx}\, dx = \frac{\lambda}{\lambda - t} = 1 + \frac{t}{\lambda} + \frac{t^2}{\lambda^2} + \cdots$$

$(t < \lambda)$

③ $X \sim$ pdf $\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, if x

$$M_X(t) = e^{t^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot e^{tx - t^2/2}\, dx = e^{t^2/2} \quad \text{if } t$$

$$= 1 + \frac{t^2}{2} + \frac{1}{2!}\frac{t^4}{2^2} + \frac{1}{3!}\frac{t^6}{2^3} + \cdots$$

Property:

$$M_X(t) = E\left[e^{tX}\right] = E\left[1 + tX + \frac{t^2X^2}{2!} + \frac{t^3X^3}{3!} + \cdots\right]$$

$M_X(0) = 1$

$M_X'(0) = E[X]$

$M_X^{(k)}(0) = E[X^k]$

$$M_X^{(k)}(t) = E\left[X^k + t(\cdots)\right]$$

X, Y : jointly RVs.

$\mathrm{Cov}(X,Y) \triangleq E[XY] - E[X]E[Y]$ $\quad \rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$ $\quad |\rho_{XY}| \le 1$

X, Y: centralized, normalized $\quad \rho_{XY} = E[XY]$

Ex: ①

$y \backslash x$	-1	0	1	
-1	$q-a$	$\frac{1}{2}-2q+a$	$q-a$	$\frac{1}{2}$
1	a	$2q-2a$	a	$\frac{1}{2}$
	q	$1-2q$	q	

$q \le \frac{1}{2}$

$E[XY] = 0$ "uncorrelated"

$1, 0, -1$

②

$y \backslash x$	-1	1	
-1	$\frac{1}{2}-a$	a	$\frac{1}{2}$
1	a	$\frac{1}{2}-a$	$\frac{1}{2}$
	$\frac{1}{2}$	$\frac{1}{2}$	

X, Y : uncorrelated

$\Updownarrow$

$aX+b,\ cY+d$: uncorrelated

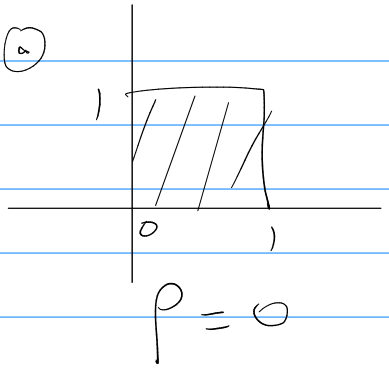
X, Y : indep

f, g: non-trivial

$\Downarrow$

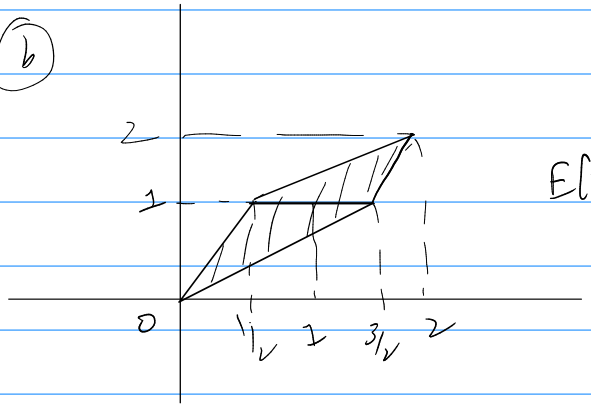
$f(X), g(Y)$: indep

③ (a)

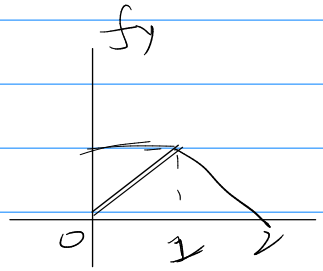


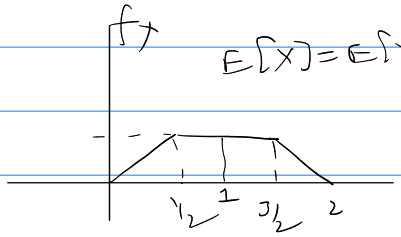
$\rho = 0$

(b)



$E[XY] \overset{?}{\ge} 1$?





$E[X] = E[Y] = 1$